Level 4

Function and Modules

Function Definitions, Arguments, Return Values, Importing Modules

Question1: Write a function that reverses a string.

```
def reverse_string(text):
    return text[::-1]

print(reverse_string("Hello, World!"))
```

```
(base) nadaabuzaid@Nadas-MacBook Level4 % python3 1.py
!dlroW ,olleH
```

Question2: Check if a password is strong (≥8 chars, includes number & special character).

```
def check_password_strength(password):
    if len(password) < 8:
        return False
    has_number = False
    has_special = False
    for char in password:
        if char.isdigit():
            has_number = True
        elif not char.isalnum():
            has_special = True
    return has_number and has_special

print(check_password_strength("Password1!"))
```

```
18  print(check_password_strength("Password1!"))

Terminal  Problems  Ports  DevDb  Debug Console  Spell Checker  Output

(base) nadaabuzaid@Nadas-MacBook Level4 % python3 2.py
True
```

Question3: Generate a random password of 12 characters.

```
import random
import string

characters = string.ascii_letters + string.digits +
string.punctuation

def generate_random_password(length=12):
    password = ""
    for i in range(length):
        password += random.choice(characters)
    return password

print(generate_random_password())
```

```
(base) nadaabuzaid@Nadas-MacBook Level4 % python3 3.py
$j?sW-o!2c1T
```

Question4: Create an MD5 hash of a user-entered string using hashlib.

```
import hashlib
entered_string = input("Enter a string to hash: ")
hashed_string =
hashlib.md5(entered_string.encode()).hexdigest()

print(f"MD5 hash of '{entered_string}' is: {hashed_string}")
```

```
(base) nadaabuzaid@Nadas-MacBook Level4 % python3 4.py
Enter a string to hash: NoBodyCanHackMe
MD5 hash of 'NoBodyCanHackMe' is: 07041830b17fb490171fabf7b7168feb
```

Question5: Check if an IP address is valid.

```
ip = input("Enter an IP address: ")
parts = ip.split(".")
if len(parts) == 4:
    valid = True
    for part in parts:
        if not part.isdigit() or not 0 <= int(part) <= 255:
            valid = False
            break
    if valid:
        print(f"{ip} is a valid IP address.")
    else:
        print(f"{ip} is not a valid IP address.")
else:
    print(f"{ip} is not a valid IP address.")
```

```
(base) nadaabuzaid@Nadas-MacBook Level4 % python3 5.py
Enter an IP address: 192.168.257.9
192.168.257.9 is not a valid IP address.
(base) nadaabuzaid@Nadas-MacBook Level4 % python3 5.py
Enter an IP address: 192.168.0.1
192.168.0.1 is a valid IP address.
```

Question6: Generate random MAC addresses.

```
import random
import string

def generate_random_mac():
    mac = [
        random.choice(string.hexdigits).upper() +
        random.choice(string.hexdigits).upper() for _ in
range(6)]
    return ":".join(mac)

print(generate_random_mac())
```

```
(base) nadaabuzaid@Nadas-MacBook Level4 % python3 6.py
A2:16:34:1E:0C:66
```

Question7: # XOR encryption function.

```
def xor_encrypt(text, key):
    encrypted = ""
    for c in text:
        encrypted_char = chr(ord(c) ^ ord(key))
        encrypted += encrypted_char
    return encrypted
```

```
(base) nadaabuzaid@Nadas-MacBook Level4 % python3 7.py
Enter text: Hello
Enter a single character key: k
Encrypted text: #
(base) nadaabuzaid@Nadas-MacBook Level4 % python3 7.py
Enter text: Nada
Enter a single character key: M
Encrypted text: ,),
```

Question8: Generate a random device ID using uuid.

```
import uuid
def generate_random_device_id():
    return str(uuid.uuid4())

print(generate_random_device_id())
```

```
(base) nadaabuzaid@Nadas-MacBook Level4 % python3 8.py
89096e5d-9aa7-4f04-951e-2ae49469168a
```

Question9: Resolve a hostname to an IP address.

```
import socket

hostname = input("Enter a hostname: ")
ip_address = socket.gethostbyname(hostname)

print("IP address:", ip_address)
```

```
(base) nadaabuzaid@Nadas-MacBook Level4 % python3 9.py
Enter a hostname: google.com
IP address: 142.250.75.46
```

Question10: Extract all vowels from a string

```
text = input("Enter a string: ")

vowels = "aeiouAEIOU"
result = ""

for char in text:
    if char in vowels:
        result += char

print("Vowels:", result)
```

```
(base) nadaabuzaid@Nadas-MacBook Level4 % python3 10.py
Enter a string: Nauedf
Vowels: aue
```